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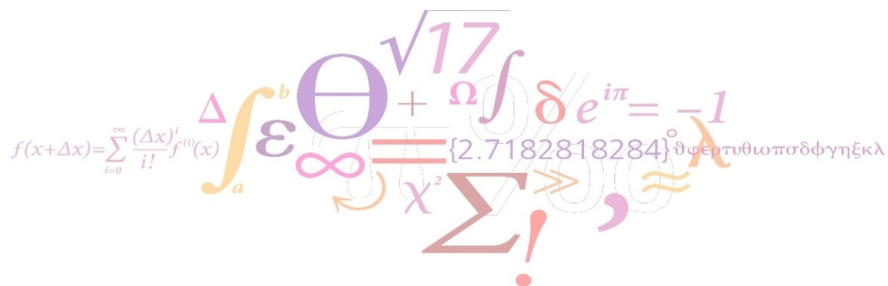


02443 Stochastic Simulation

Project 2 - Simulation of Epidemics

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1 Part I: Basic Modelling

1.1 Introduction of the basic SIR model

The classical SIR model (based on [1]) simplifies the transmission dynamics of infectious diseases by grouping a fixed population of size N into three mutually exclusive compartments $S(t)$, $I(t)$, and $R(t)$:

- **$S(t)$ (Susceptible):** Number of healthy individuals with no prior immunity who are susceptible to infection.
- **$I(t)$ (Infectious):** Number of infected individuals who can transmit the disease to susceptible individuals.
- **$R(t)$ (Recovered):** Number of individuals who have recovered from infection and obtained permanent immunity.

At any given time t , the total population is kept constant:

$$S(t) + I(t) + R(t) = N. \quad (1)$$

The standard baseline SIR model relies on the following assumptions:

- **Homogeneous mixing:** Every individual has an equal probability of contacting any other individual, ignoring spatial distribution, age groups, or social networks.
- **Constant population:** The epidemic occurs over a short timescale such that natural births and unrelated background deaths are negligible (N remains constant).
- **No incubation period:** Transmission leads to immediate contagiousness. There is no latent or exposed period (E) before an individual becomes infectious.
- **Permanent immunity:** Recovered individuals retain life-long immunity. Re-infection is impossible.

We will investigate the epidemic dynamics through two different frameworks: deterministic and stochastic. The deterministic model is formulated as a system of ordinary differential equations (ODEs). The stochastic model is based on continuous-time Markov chain (CTMC) simulated using a standard event-driven simulation algorithm.

Throughout the report we will use the notation: $S = S(t)$, $I = I(t)$, $R = R(t)$.

1.1.1 The Deterministic Framework (ODEs)

The deterministic model is described by the following system of ordinary differential equations (ODEs):

$$\begin{aligned} \frac{dS}{dt} &= -\frac{\beta SI}{N}, & \frac{dI}{dt} &= \frac{\beta SI}{N} - \gamma I, \\ \frac{dR}{dt} &= \gamma I, & \frac{dN}{dt} &= 0. \end{aligned} \quad (2)$$

where β is the infection rate and γ is the recovery rate.

1.1.2 The Stochastic Framework

In the stochastic model, the epidemic is described as a continuous-time Markov chain (CTMC), where the population changes through discrete events. For the basic SIR model, there are two possible events: infection and recovery. The corresponding state transitions and transition rates are shown in Table 1.

Table 1: Events and transition rates for the basic SIR model.

Event	Transition	Rate
Getting infected	$S \rightarrow S - 1, \quad I \rightarrow I + 1$	$\frac{\beta}{N}SI$
Recovering	$I \rightarrow I - 1, \quad R \rightarrow R + 1$	γI

The CTMC is simulated by repeatedly sampling the waiting time until the next event and updating the system state. At each step, the total transition rate a_0 is computed as the sum of all possible transition rates. For the basic SIR model that becomes:

$$a_0 = \frac{\beta}{N}SI + \gamma I. \quad (3)$$

The waiting time until the next event is sampled from an exponential distribution with rate parameter a_0 . Finally, the type of event is selected with probability proportional to its transition rate.

Unless otherwise stated, the CTMC results are based on 5 independent realizations. 5 runs were chosen to showcase the stochastic variability, while keeping the computational cost reasonable.

1.1.3 Model Parameters

The dynamics of the SIR model are determined by two main parameters:

- **β (Transmission rate):** The average number of successful, infection-transmitting contacts made by a single infectious individual per unit of time.
- **γ (Recovery rate):** The rate at which individuals recover from the infection. The average duration of the infectious period is mathematically defined as $1/\gamma$.

An important quantity in epidemiology is the *basic reproduction number*, defined as the dimensionless ratio:

$$R_0 = \frac{\beta}{\gamma} \quad (4)$$

It represents the expected number of secondary infections caused by one infected individual in a fully susceptible population ($S \approx N$).

- **$R_0 < 1$:** Each infected individual infects fewer than one new person on average. The disease cannot sustain itself and will die out without causing an epidemic.

- $R_0 > 1$: Each case reproduces more than one new case. The system exhibits initial exponential growth, leading to an epidemic outbreak.

Table 2 presents historical literature estimates for parameter values of the three viruses we will investigate throughout this report.

Table 2: Estimated parameter values for three pathogens.

Pathogen	R_0	γ (day ⁻¹)	β (day ⁻¹)	Source
Seasonal Influenza	1.5	0.33	0.50	[2]
COVID-19 (Wuhan)	2.5	0.10	0.25	[3]
Ebola Virus	1.5	0.10	0.15	[4]

1.2 Simulation and Comparative Analysis of Population Sizes

To evaluate the precision of the deterministic model against the stochastic formulation, we simulate a standard outbreak scenario over a time window of $t_{\max} = 100$ days (where one time unit equals one day). The models are simulated using $\beta = 0.5$ day⁻¹ and $\gamma = 0.25$ day⁻¹, and are compared across three distinct population scales: small ($N = 100$), medium ($N = 10\,000$), and large ($N = 1\,000\,000$).

To eliminate any artifact delays caused by initial dilution and ensure a mathematically fair baseline across scales, the initial infectious pool is set proportionally to 5% of the population ($I_0 = 0.05N$, with $S_0 = 0.95N$). The stochastic CTMC is simulated using 5 independent realizations per population size via the algorithm described in Section 1.1.2.

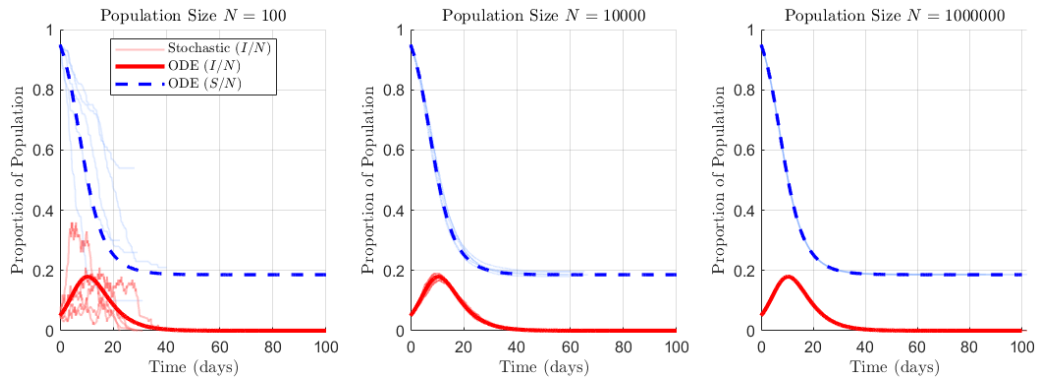


Figure 1: Comparison of Stochastic Realizations and Deterministic ODEs across various population sizes.

The simulation results, captured in Figure 1, yield the following key insights:

- **Small Population ($N = 100$):** The deterministic framework provides a poor representation of individual real-world outcomes. At this scale, demographic stochasticity dominates the system. At this scale, the paths display highly jagged trajectories and significant phase shifts compared to the ODE baseline.

- **Medium Population** ($N = 10\,000$): The stochastic realizations begin to tightly bundle around the deterministic trend. While individual variations in peak timing and magnitude are still visibly present as thin fluctuating bands, the global trajectory shape conforms well to the underlying continuous averages.
- **Large Population** ($N = 1\,000\,000$): The random noise of individual transmission events completely averages out due to the Law of Large Numbers. All 5 independent stochastic trajectories converge perfectly, becoming visually indistinguishable from the smooth deterministic ODE curves.

1.3 Comparative Analysis of Pathogen Profiles

To study the structural consequences of distinct epidemiological parameters, a deterministic simulation was conducted on a large, stable population ($N = 1\,000\,000$) over a 150-day period. The initial conditions were set identically across all trials with a proportional infection load of 5% ($I_0 = 0.05N$). Three different pathogens were evaluated based on literature-derived constants: Seasonal Influenza ($\beta = 0.50, \gamma = 0.33$), COVID-19 Wuhan ancestral strain ($\beta = 0.25, \gamma = 0.10$), and Ebola Virus ($\beta = 0.15, \gamma = 0.10$).

The global comparison of the infection trajectories is illustrated in Figure 2, while the complete individual compartment timelines (S, I, R) for each virus are detailed in Figure 3.

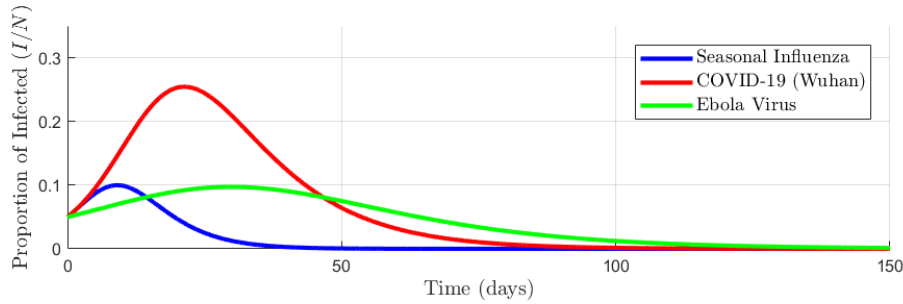


Figure 2: Epidemic Curve Comparison ($N = 1\,000\,000$, $I_0 = 5\%$)

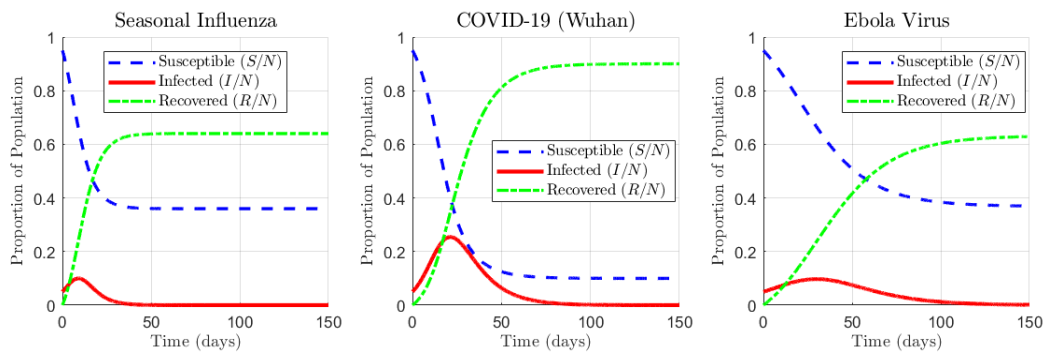


Figure 3: Individual Compartment Trajectories for Multi-Pathogen SIR Dynamics

The simulation results reveal how the basic reproduction number ($R_0 = \beta/\gamma$) and the transmission rate (β) dictate the scale, severity, and timing of an outbreak:

- **Seasonal Influenza:** Characterized by a moderate basic reproduction number ($R_0 \approx 1.51$), the virus presents a classic, sharp epidemic curve. Driven by a high recovery rate ($\gamma = 0.33$), which corresponds to a short infectious window ($1/\gamma \approx 3$ days), the infection peaks rapidly around day 10 at approximately 10% of the population. The outbreak completely burns out by day 40, leaving roughly 36% of the population uninfected ($S/N \approx 0.36$).
- **COVID-19 (Wuhan):** Exhibiting an intermediate reproduction profile ($R_0 = 2.5$) combined with a low recovery rate ($\gamma = 0.10$, yielding a prolonged 10-day infectious window), this pathogen produces a massive epidemic wave. The infection curve peaks around day 22, reaching a high maximum intensity where over 25% of the entire population is actively infected simultaneously. Consequently, the susceptible pool is severely depleted, with roughly 90% of the population eventually moving into the recovered/immune pool ($R/N \approx 0.90$) by day 75.
- **Ebola Virus:** Representing a pathogen with a low transmission rate ($\beta = 0.15$) relative to its clinical persistence ($\gamma = 0.10$), its reproductive threshold sits closely to influenza ($R_0 = 1.5$). However, because its transmission rate is low, the wave is significantly flattened and drawn out. It scales gradually from its 5% baseline to a late peak around day 36, maxing out just below 10%. The system takes over 120 days to fully stabilize, showcasing a long, lingering epidemiological footprint.

1.4 Likelihood of Disease Disappearance

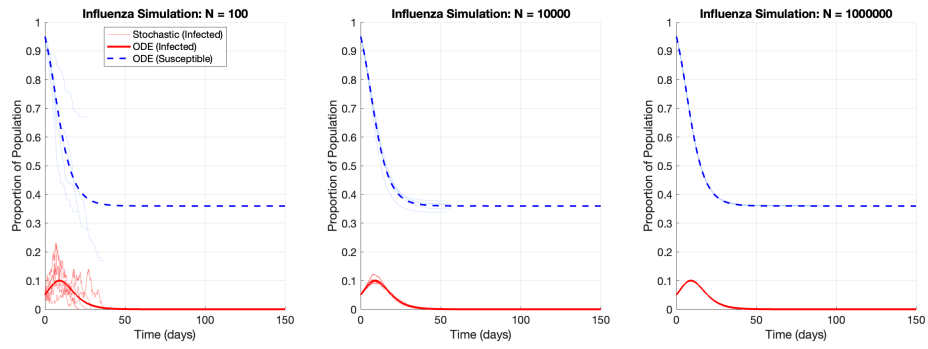
In a closed SIR model, the disease eventually dies out because recovered individuals gain immunity and cannot be infected again. As the disease spreads, the number of susceptible individuals decreases, which makes it harder for transmission to continue. Eventually, this leads to the number of infected individuals going to zero.

Figure 4 shows simulation results for seasonal influenza, COVID-19, and Ebola for different population sizes. For all three diseases, the likelihood of disease disappearance depends strongly on the population size.

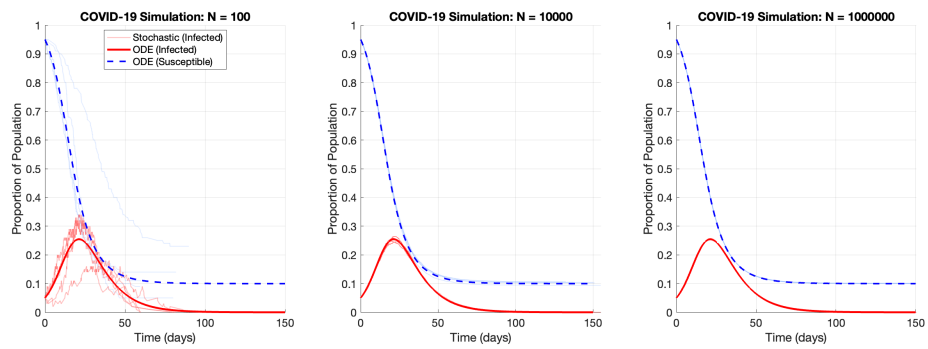
For $N = 100$, many of the stochastic simulations die out shortly after the outbreak begins. This is especially visible for seasonal influenza and Ebola, where several trajectories quickly reach zero infected individuals. Since only a small number of individuals are infected at the beginning, random fluctuations can have a large effect on whether the outbreak develops or not. In some cases, infected individuals recover before infecting others, which leads to early extinction.

As the population size increases, the stochastic simulations become more similar to the deterministic ODE solution. This can be seen for all three diseases. For $N = 10^6$, the stochastic trajectories are almost identical to the deterministic curve, meaning that random fluctuations have little effect on the overall dynamics. In this case, the disease typically persists until the susceptible population becomes too small to sustain transmission.

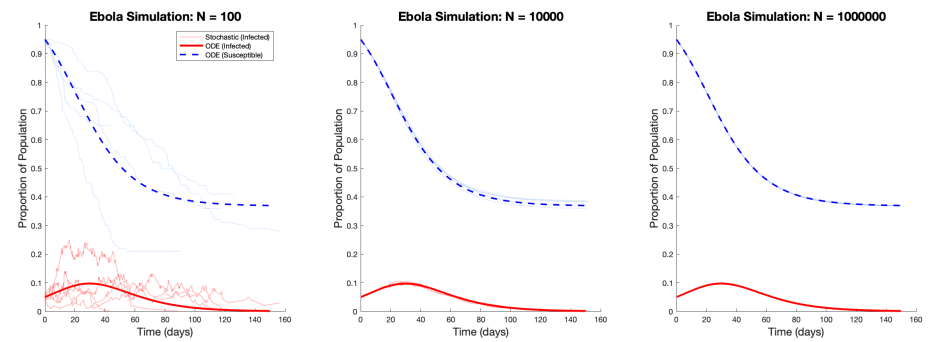
Overall, the results show that disease extinction due to stochastic effects is much more likely in small populations. Even though all outbreaks eventually die out in the closed SIR model, small populations often experience early stochastic extinction before a large epidemic develops.



(a) Seasonal influenza



(b) COVID-19



(c) Ebola

Figure 4: Simulation trajectories showing stochastic extinction versus deterministic burnout across three pathogens for varying population sizes N .

2 Part II: Extended Modelling

2.1 Including Birth and Death Rates

We now extend the basic model from Part 1 (Section 1) to include birth and death rates. The population size is no longer constant and throughout this subsection we use the notation $N = N(t) = S(t) + I(t) + R(t)$, with the initial population size being $N_0 = N(0)$. The deterministic model (ODEs) become (based on [1]):

$$\begin{aligned}\frac{dS}{dt} &= \Lambda - \mu S - \frac{\beta}{N}IS, \\ \frac{dI}{dt} &= \frac{\beta}{N}IS - \gamma I - \mu I - \alpha I, \\ \frac{dR}{dt} &= \gamma I - \mu R,\end{aligned}\tag{5}$$

where Λ is the birth rate, μ is the general death rate, and α is the death rate caused directly by the disease. All rates are per time unit, i.e. per day. We assume that the number of births is equal to the number of deaths not caused by the disease:

$$\Lambda = \mu S + \mu I + \mu R = \mu N,\tag{6}$$

which results in the population being constant in the absence of disease-induced deaths.

α is computed based on the fatality probability p_F , i.e. the probability of dying because of the disease:

$$p_F = P(\text{disease-induced death} \mid \text{infected}) = \frac{\alpha}{\alpha + \gamma + \mu} \quad \Leftrightarrow \quad \alpha = \frac{p_F}{1 - p_F}(\gamma + \mu).\tag{7}$$

We assume a non-disease-related death rate of $\mu = 2.7 \cdot 10^{-5} \text{ day}^{-1}$ [5]. The assumed values of the other new parameters introduced can be found in Table 3. In the stochastic CTMC model (described in Section 1.1.2) there now exist 6 events, which can be seen in Table 4, together with the corresponding transition rates.

Table 3: Assumed values of fatality probability p_F and disease-induced death rate α for the three viral diseases.

Pathogen	p_F	α (day ⁻¹)	Source
Seasonal influenza	0.0001	$3.3 \cdot 10^{-5}$	[6]
COVID-19	0.003	$3.0 \cdot 10^{-4}$	[7]
Ebola	0.5	$1.0 \cdot 10^{-1}$	[8]

Figure 5 shows the results of the model over 150 days. Seasonal influenza and COVID-19 are only mildly affected by the inclusion of birth and death rates, as their dynamics strongly resemble those observed in Figure 3. In contrast, Ebola is strongly affected due to the high disease-induced death rate. The number of infected individuals quickly decreases, causing the disease to “die out” quickly. Consequently, only a small fraction of the population is recovered individuals, since many infected individuals die before recovering. It shows that

Table 4: Events and transition rates for SIR model with birth and death rates included.

Event	Transition	Rate
Getting infected	$S \rightarrow S - 1, \quad I \rightarrow I + 1$	$\frac{\beta}{N}SI$
Recovering	$I \rightarrow I - 1, \quad R \rightarrow R + 1$	γI
Birth	$S \rightarrow S + 1, \quad N \rightarrow N + 1$	$\Lambda = \mu N$
Dying while susceptible	$S \rightarrow S - 1, \quad N \rightarrow N - 1$	μS
Dying while infected	$I \rightarrow I - 1, \quad N \rightarrow N - 1$	$\mu I + \alpha I$
Dying while recovered	$R \rightarrow R - 1, \quad N \rightarrow N - 1$	μR

even a highly deadly disease does not necessarily extinguish the population, since the ability to spread the disease is diminished due to infected individuals dying before successfully spreading it. Overall, the results illustrate that including especially disease-induced deaths can significantly change the epidemic dynamics, if the disease-induced death rate is sufficiently large.

Since birth and death rates have been included, the total population size N becomes time-dependent. Figure 6 shows the evolution of N over the 150-day simulation period. The population remains nearly constant for seasonal influenza and COVID-19, whereas Ebola causes a reduction of roughly 800 individuals over the first 50 days due to its high disease-induced death rate.

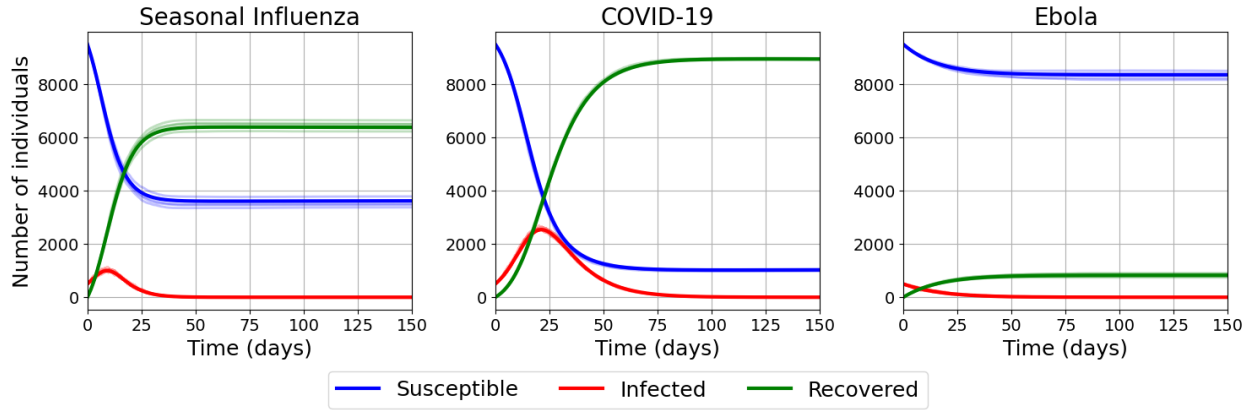


Figure 5: SIR model with birth and death rates included. Bold lines are ODE solutions, while thin lines are CTMC realizations. The close agreement between the models causes many of the CTMC realizations to overlap the ODE solutions. Initial population size $N_0 = 10\,000$, with initial number of infected at 5%.

2.1.1 Can a highly deadly disease extinguish the population?

We dive a bit deeper into whether a highly deadly disease can extinguish the population or not. We consider three very extreme (and unrealistic) parameter settings, with various degrees of high transmission rate, low recovery rate, and high probability of disease-induced death. The results can be seen in Figure 7 and 8. Summarized:

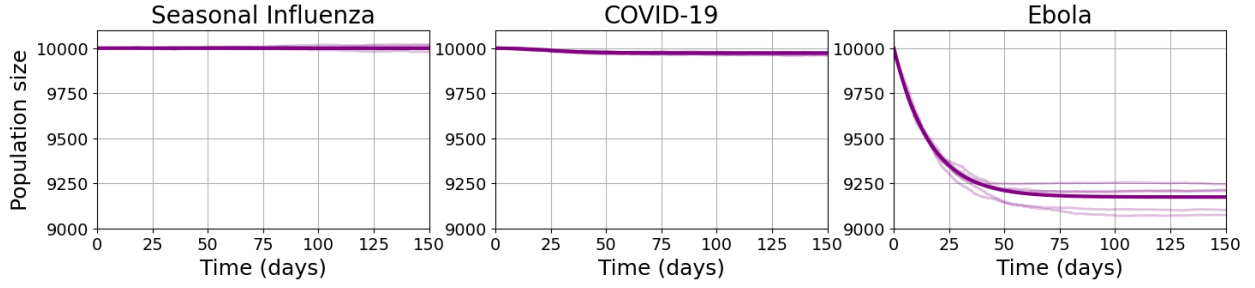


Figure 6: Total population size as a function of time for SIR model with birth and death rates included.

- Leftmost case: Almost everyone is infected very quickly, but some still recover. The population size is greatly reduced but stable and not extinguished.
- Middle case: Almost everyone is infected almost immediately and since the recovery rate is so low, almost everyone dies. However, the population never reaches 0 in any of the 5 CTMC simulations over the 150-day period.
- Rightmost case: The disease-induced death rate is so high that all of the infected die too quickly to spread the disease. Thus, most of the population survives.

In none of the three extreme cases does the population size reach zero and becomes extinct. However, the population size of the middle case reaches on average an extremely small population size of $N(150) = 7$ across the 5 CTMC simulations. Since $I(150) = 0$ for all 5 simulations, the disease will not spread further, and extending the simulations to 50 000 days shows that the remaining population size stays approximately constant. In reality, such a small population size will probably lead to problems with inbreeding, which is not considered in our model. In conclusion, these results indicate that, even for the extreme parameter settings considered here, a highly deadly disease does not necessarily lead to extinction of the population.

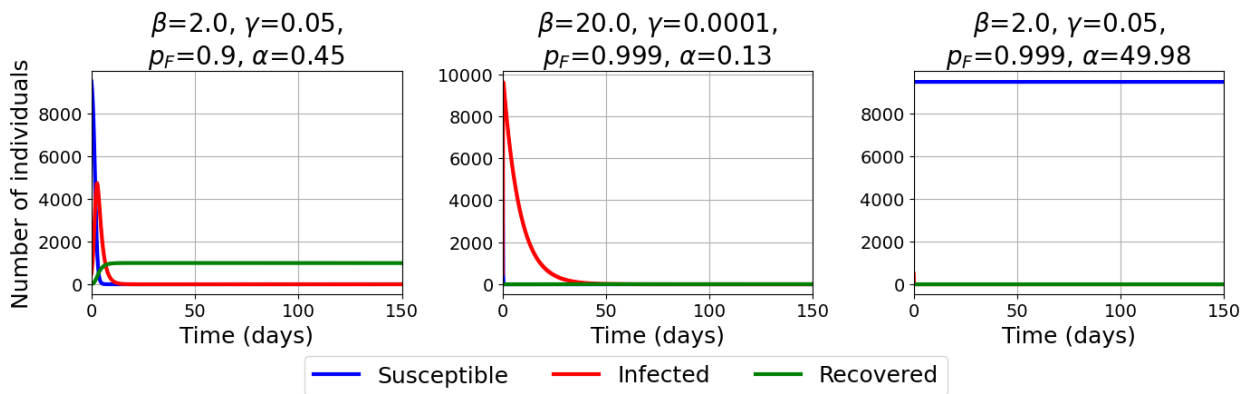


Figure 7: SIR model with birth and death rates included using extreme parameter settings. Initial population size $N_0 = 10\,000$, with initial number of infected at 5%.

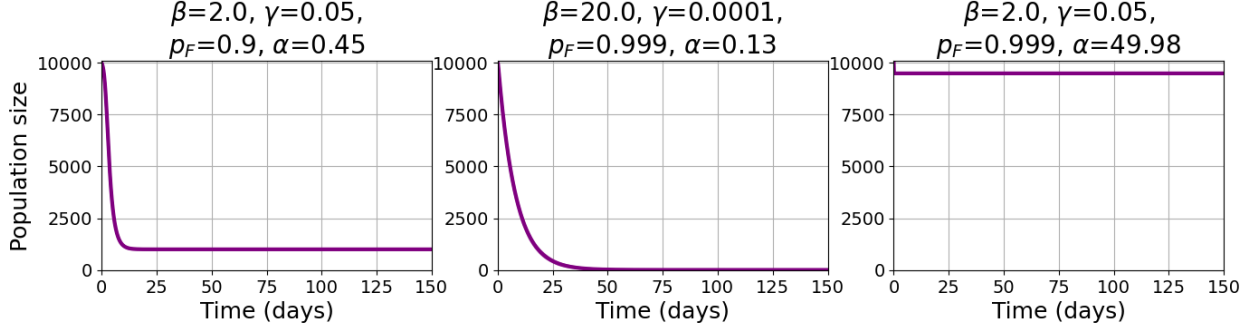


Figure 8: Total population size as a function of time for SIR model with birth and death rates included using extreme parameter settings.

2.2 SEIR Model Based on COVID-19

We extend the basic framework from Section 1 by introducing a pre-symptomatic infectious phase. The deterministic system is described by the following ordinary differential equations:

$$\begin{aligned}
 \frac{dS}{dt} &= -\frac{\beta_E E + \beta_I I}{N} S, \\
 \frac{dE}{dt} &= \frac{\beta_E E + \beta_I I}{N} S - \sigma E, \\
 \frac{dI}{dt} &= \sigma E - \gamma I, \\
 \frac{dR}{dt} &= \gamma I.
 \end{aligned} \tag{8}$$

where β_E and β_I are the daily transmission rates for pre-symptomatic and symptomatic individuals, σ is the rate of symptom onset, and γ is the recovery rate. The average durations of the pre-symptomatic and symptomatic infectious periods are given by $1/\sigma$ and $1/\gamma$, respectively.

The corresponding stochastic model follows the events and transition rates outlined in Table 5. This formulation splits the force of infection into two distinct pathways to account for transmission originating from both compartments.

Table 5: Events and transition rates for the extended SEIR model.

Event	Transition	Rate
Infection	$S \rightarrow S - 1, \quad E \rightarrow E + 1$	$\frac{\beta_E}{N} SE + \frac{\beta_I}{N} SI$
Symptom onset	$E \rightarrow E - 1, \quad I \rightarrow I + 1$	σE
Recovery	$I \rightarrow I - 1, \quad R \rightarrow R + 1$	γI

Figure 9 displays the simulated trajectories across the three population scales. For the smallest population size ($N = 100$), demographic noise dominates; the discrete paths diverge substantially from the ODE baseline, with several realizations experiencing early stochastic

extinction, while surviving outbreaks display high variance in peak timing and magnitude. As the scale increases to $N = 1\,000\,000$, these relative random fluctuations damp out at a rate of $1/\sqrt{N}$. The stochastic paths smooth out and converge onto the deterministic solution, as expected via the Law of Large Numbers.

The plots also reveal key properties of the underlying transmission structure. There is a distinct phase lag between the two infected states, where the pre-symptomatic compartment (E) peaks several days before the symptomatic compartment (I). This horizontal shift matches the biological incubation delay determined by $1/\sigma$. Furthermore, because transmission occurs during this pre-symptomatic window (β_E), a significant portion of the outbreak spreads silently before a visible pool of symptomatic cases emerges. This dynamic shows why isolation strategies based solely on active symptoms are less effective when substantial transmission occurs before symptoms appear.

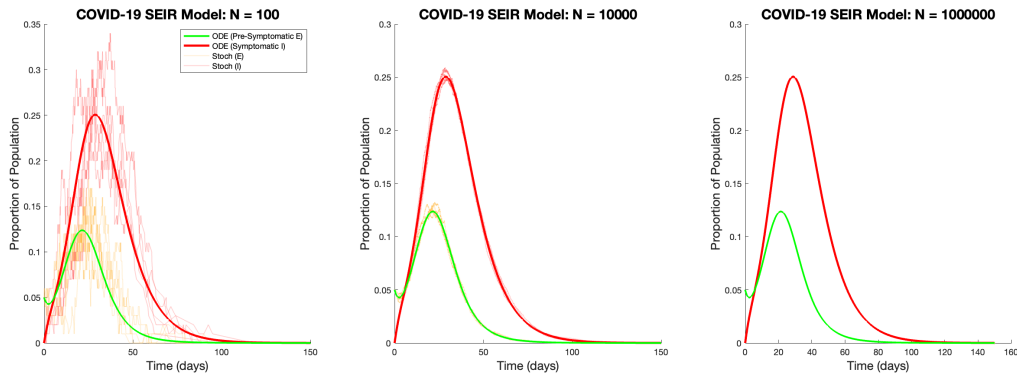


Figure 9: Deterministic versus stochastic realizations of the extended pre-symptomatic SEIR model for COVID-19 across scales $N = 100$, $N = 10\,000$, and $N = 1\,000\,000$.

2.3 Effect of Vaccination on COVID-19

Here, we will investigate the effect of vaccination on COVID-19. We extend the basic model (Section 1) to include a fourth compartment V of successfully vaccinated individuals. The deterministic model of ODEs is updated to be (inspired by [1]):

$$\begin{aligned} \frac{dS}{dt} &= -\frac{\beta}{N}IS - \rho eS, \\ \frac{dI}{dt} &= \frac{\beta}{N}IS - \gamma I, \\ \frac{dR}{dt} &= \gamma I, \\ \frac{dV}{dt} &= \rho eS, \end{aligned} \tag{9}$$

where ρ is the vaccination rate and $e \in [0, 1]$ is the effectiveness of the vaccine. Thus, an individual receiving the vaccine has probability e of becoming immune and moving to the vaccinated compartment. The model assumes that only susceptible individuals can be vaccinated, that vaccinations start immediately and continue forever, and that successfully vaccinated individuals obtain permanent immunity.

The events and transition rates of the stochastic model can be seen in Table 6. We assume that $\rho = 0.05 \text{ day}^{-1}$, and test different values for the vaccine effectiveness $e \in \{0.3, 0.6, 0.9\}$ (loosely inspired by [9]).

Table 6: Events and transition rates for SIR model with vaccination included.

Event	Transition	Rate
Getting infected	$S \rightarrow S - 1, \quad I \rightarrow I + 1$	$\frac{\beta}{N}SI$
Recovering	$I \rightarrow I - 1, \quad R \rightarrow R + 1$	γI
Successfully vaccinated	$S \rightarrow S - 1, \quad V \rightarrow V + 1$	ρeS

The results can be seen in Figure 10. As the vaccine effectiveness increases, the epidemic peak decreases. The epidemic peak decreases from approximately 2 550 infected individuals without vaccination to 1 100 infected individuals for a vaccine effectiveness of 90%, corresponding to a reduction of about 57%. This occurs because susceptible individuals are moved to the vaccinated compartment before they get infected. As a result, fewer individuals need to become immune through infection and subsequent recovery, since they get immunity through vaccination instead. Furthermore, as the vaccine effectiveness increases, the outbreak duration seems to decrease. Lastly, even a high vaccine effectiveness ($e = 0.9$) does not completely eliminate the disease, but reduces the total number of infections.

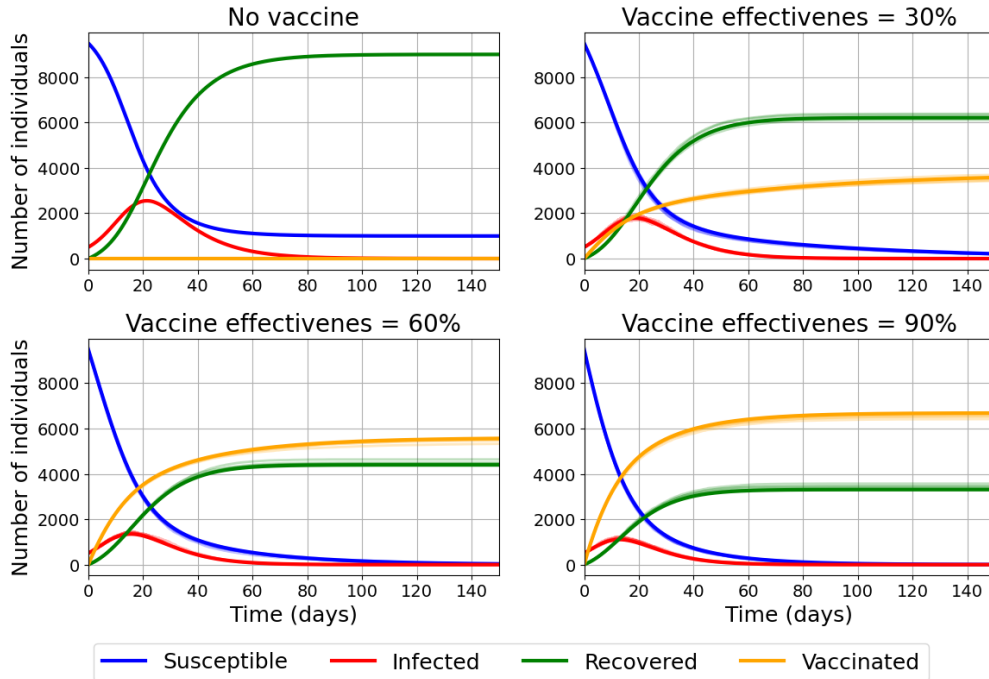


Figure 10: SIR model with vaccination included for COVID-19. Bold lines are ODE solutions, while thin lines are CTMC realizations. The close agreement between the models causes many of the CTMC realizations to overlap the ODE solutions. Population size $N = 10\,000$, with initial number of infected at 5%.

2.4 Spatial Distribution with State-Dependent Movement

As in the other extensions of Part II, we start from the basic SIR model of Part I (Section 1). Throughout this subsection we will refer to the basic SIR model as the *well-mixed model*, since it assumes homogeneous mixing, where every individual can infect any other individual with equal probability. In this part we relax that assumption and give each person a location, so that infection is local and the movement of a person can depend on its disease state.

We place N individuals on an $L \times L$ grid. The edges of the grid wrap around, so every cell has the same eight neighbours and no individual sits on a boundary. Each individual has a position on the grid and a compartment S , I or R . The neighbourhood of an individual is its own cell and the eight cells around it. We keep the CTMC and the Gillespie direct method from Part I, and define three types of events, each with a rate.

A susceptible individual whose neighbourhood holds I_h infected out of N_h individuals gets infected at rate $\beta I_h/N_h$. Adding this over all susceptibles, the total infection rate is

$$a_{\text{inf}} = \beta \sum_c S_c \frac{I_{h,c}}{N_{h,c}}, \quad (10)$$

where the sum runs over grid cells, S_c is the number of susceptibles in cell c , and $I_{h,c}$ and $N_{h,c}$ are the infected count and the total count in the neighbourhood of cell c . Each infected individual recovers at rate γ , so the total recovery rate is $a_{\text{rec}} = \gamma I$, the same as in Part I. Each individual also moves to a random neighbouring cell at a rate that depends on its state. Healthy individuals (S and R) move at rate m_{heal} and infected individuals move at the lower rate m_{inf} . Moving only changes positions, and this is how the movement depends on the state. Table 7 summarises the three events and their rates.

Table 7: Events and transition rates for the spatial SIR model with state-dependent movement.

Event	Transition	Rate
Getting infected	$S \rightarrow S - 1, \quad I \rightarrow I + 1$ (local)	$\beta I_h/N_h$ per susceptible
Recovering	$I \rightarrow I - 1, \quad R \rightarrow R + 1$	γI
Moving (healthy)	Individual moves to a neighbouring cell	m_{heal} per S or R
Moving (infected)	Individual moves to a neighbouring cell	m_{inf} per I

We use the three diseases in Table 2 with their β and γ values. We take $N = 1000$ individuals on an $L = 30$ grid, which is about one person per cell. This is small enough to run quickly and average over several runs, while still showing the local structure. The 5% initial infected are placed as one cluster in the centre, so we can follow the spread from a single clear starting point instead of from infected scattered everywhere. We set $m_{\text{heal}} = 0.5$ and $m_{\text{inf}} = 0.1$ per day, so infected individuals move five times less than healthy ones, since sick people tend to stay put. These movement rates are modelling choices and are not taken from the literature. All runs use the seed `np.random.default_rng(42)`, and curves that we compare across models are averaged over 5 runs.

With the neighbourhood set to the whole grid, the spatial model is the same as the well-mixed model, so the two should agree. Figure 11 shows that the spatial model with global mixing, the well-mixed CTMC, and the deterministic ODE all lie on top of each other for COVID-19. This tells us the code is correct, so any difference we see later comes from the local spatial structure and not from a bug.

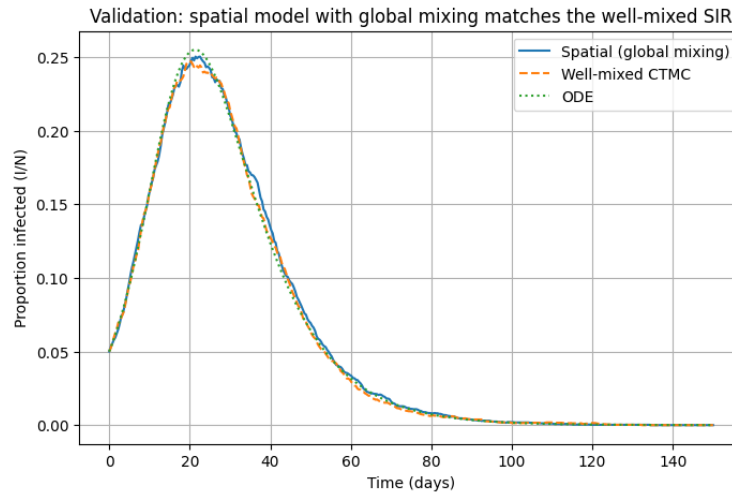


Figure 11: Validation of the spatial model. With global mixing the spatial model reproduces the well-mixed CTMC and the deterministic ODE for COVID-19 ($N = 1000$, mean of 5 runs).

Figure 12 follows a COVID-19 outbreak on the grid. At the start all the infected sit in a small cluster in the centre and the rest of the grid is susceptible. After 30 days the infection has spread to a ring around the seed/starting point, and the centre has already turned to recovered, so the infected form a band between the recovered core and the susceptibles outside. After 60 days this band has moved further out and the recovered region in the middle is much larger. The infection never appears far from where it already is. It travels as a front that uses up the susceptibles at its edge and leaves recovered individuals behind it. This is the spatial picture that the well-mixed model cannot show, because there every individual mixes with the whole population at once. We show COVID-19 here because its higher R_0 produces a clearly visible front, while for influenza and Ebola the spatial outbreak barely spreads, as Figure 13 shows.

Figure 13 compares the spatial and well-mixed dynamics for the three diseases. In every case the spatial epidemic peaks much lower and later than the well-mixed one, and a large part of the population is never infected. For COVID-19 the well-mixed susceptible curve falls to about 0.1, while in the spatial model it stays near 0.8, so most individuals escape the disease. The reason is that infection can only pass between near neighbours. The susceptibles next to the front are used up quickly, but once they are gone the local supply of new cases runs out, while susceptibles further away are never reached. The epidemic therefore spreads only as fast as the front can move, instead of taking off everywhere at once, which makes the curve lower and more drawn out. The effect is strongest for influenza and Ebola, which have the lowest $R_0 = 1.5$. In the well-mixed model these already give small outbreaks, and in the spatial model the front can barely keep itself going, so the infection almost dies out from

the start. Adding space therefore slows the real spread of the disease below what β and γ alone would suggest.

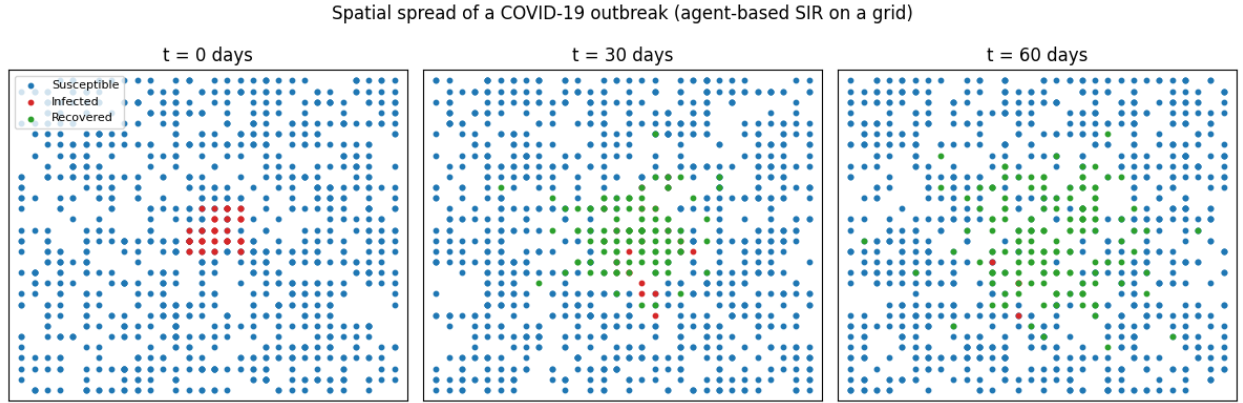


Figure 12: Spatial spread of a COVID-19 outbreak. Susceptible (blue), infected (red) and recovered (green) individuals on the grid at three times. The outbreak grows as a front from the central seed.

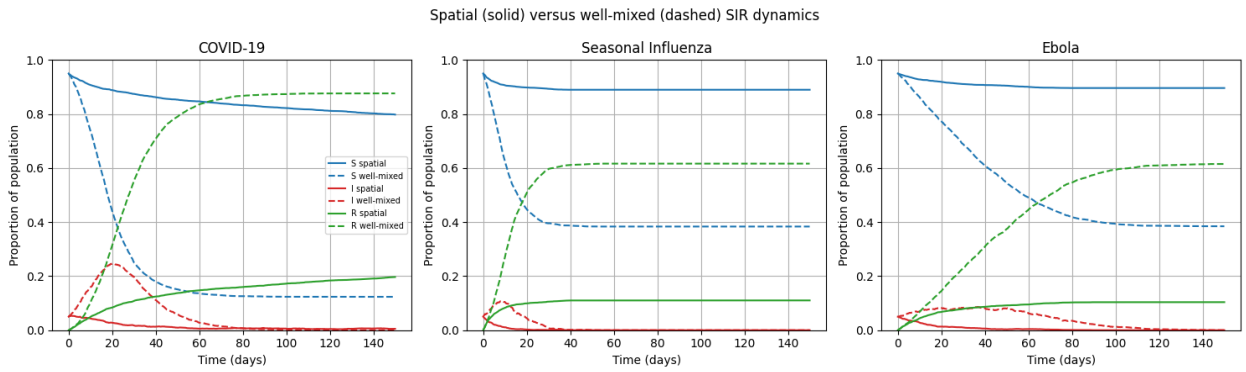


Figure 13: Spatial (solid) versus well-mixed (dashed) SIR dynamics for the three diseases ($N = 1000$, mean of 5 runs). Local contacts delay and flatten the epidemic peak.

Figure 14 shows the effect of the movement rule for COVID-19. It compares infected who move less ($m_{\text{inf}} = 0.1$) with infected who move normally ($m_{\text{inf}} = 0.5$), with everything else kept the same. When infected individuals move normally they keep meeting fresh susceptibles in new parts of the grid, so they act like moving seeds and the infection stays near its starting level for a long time. When they move less they stay surrounded by neighbours who are already infected or recovered, the local susceptibles run out, and the infection dies down. So how the infected move, and not only the contact rate β , decides whether the spatial epidemic keeps going or fades.

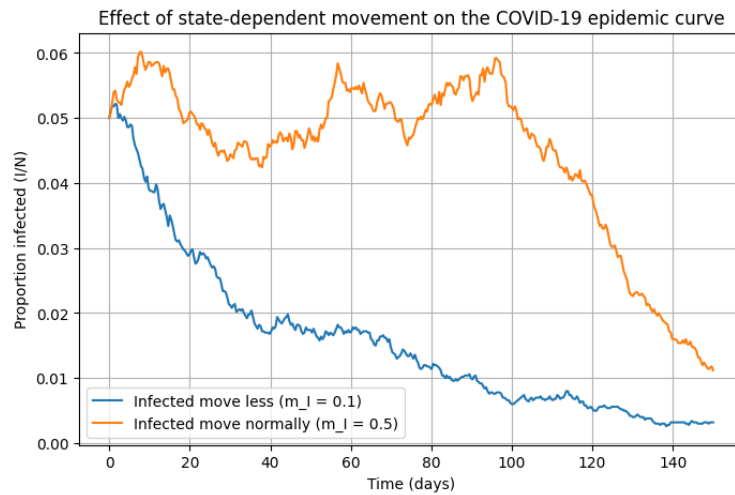


Figure 14: Effect of state-dependent movement on the COVID-19 epidemic curve. Reduced mobility of infected individuals lowers and delays the peak.

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